



CMPG 325 — Computer Networks

Milestone 2: Network Implementation Report

Secure Network Implementation with SSH Device Management

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Assigned challenge	SSH — secure device management plane
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Table of Contents

1. Executive Summary	2
2. Introduction	1-2
2.1 Recap of Milestone 1	1
2.2 Purpose of Milestone 2	2
2.3 Structure of this Report	2
3. Network Implementation Overview	2-7
3.1 Final Topology	2
3.2 VLAN Allocation	3
3.3 IP Addressing	3
3.4 Trunk Configuration	4
3.5 Inter-VLAN Routing and SVI Configuration	4
3.6 Management Plane Addressing	6
3.7 Router Configuration (R1)	7
4. Assigned Networking Feature: SSH	7-9
4.1 Design Approach	7
4.2 Configuration Applied	7
4.3 Key Decisions	7-8
4.4 Verification	8-9
4.5 Telnet Rejection	9
5. Testing and Verification	10-12
5.1 Inter-VLAN Connectivity Tests	10-11
5.2 SSH Tests	11
5.3 Trunk Tests	11-12
6. Interpretation of Results	12
7. Troubleshooting Log	13
8. Conclusion	13
9. References	14

1.Executive Summary

This report presents the implementation of the Molemane Community Newspaper network based on the design developed in Milestone 1. The network was implemented and configured using Cisco Packet Tracer and consists of one internal boundary router (R1), one Layer 3 core switch (SW-CORE), and five access switches supporting the Administration, Editorial, Sales, Design, and Boardroom departments. The network also includes a file, print and DNS server, as well as a dedicated management PC.

A total of eight VLANs were configured, including four departmental VLANs, a Boardroom VLAN, a Server VLAN, a reserved Voice VLAN for future client change request CR11, and Management VLAN 99. Inter-VLAN communication is handled by the SVIs on SW-CORE, following the VLSM addressing scheme developed in Milestone 1 using the allocated 172.30.84.0/23 address block. All connections between switches use 802.1Q trunking, with VLAN 99 configured as the native VLAN.

The required networking feature, Secure Shell (SSH), was implemented across all seven network devices, consisting of R1 and the six switches. SSH version 2 was configured together with a local administrator account and 2048-bit RSA encryption keys. The virtual terminal lines were also configured to allow SSH connections only. Management communication is restricted to VLAN 99, which uses the 172.30.84.192/28 subnet.

Network testing showed that PC-ADMIN can communicate with devices in Editorial, Sales, Design, Server, and Boardroom VLANs. Testing also verified that SSH version 2 is active on SW-CORE, SSH connections can be established from PC-MANAGEMENT and R1, and Telnet access is not permitted. Nine different categories of problems were encountered and resolved during the implementation, with the details documented in the troubleshooting log in Section 7. Overall, the implemented network satisfies Milestone 2 requirements and is ready for final submission.

2. Introduction

2.1 Recap of Milestone 1

Milestone 1 focused on reviewing and finalizing the client's network design requirements. It resulted in the development of the client requirements, physical topology, logical VLAN topology, and a VLSM-based IP addressing plan using the allocated 172.30.84.0/23 address block. An initial GitHub repository was also created. The design identified R1 as an internal boundary router with no connection to the Internet or an ISP, intended only for supporting inter-VLAN communication and SSH-based network management. It also included a Boardroom area with both wired and wireless presentation access, along with a reserved voice VLAN to support the client's CR11 request for future VoIP handsets for management.

2.2 Purpose of Milestone 2

The purpose of Milestone 2 is to review the network implementation. The design developed in Milestone 1 is converted into a functional Packet Tracer configuration, with the assigned SSH challenge implemented as a secure management mechanism. The completed network is then validated through documented testing. Any differences between the original Milestone 1 design and the final implementation are clearly identified and explained.

2.3 Structure of this Report

Section 3 outlines the completed network implementation, covering the topology, VLANs, trunk connections, SVIs, and router configuration. Section 4 explains the SSH feature, including how it was configured and the reasoning behind the chosen approach. Section 5 presents the testing and verification performed on the network. Section 6 discusses and interprets the test results, while Section 7 documents the troubleshooting process and issues resolved. Finally, Section 8 provides a conclusion and evaluates the implementation against the client's requirements.

3. Network Implementation Overview

3.1 Final Topology

The final topology follows the physical design established in Milestone 1. R1 is connected to SW-CORE, which functions as the Layer 3 core switch and handles all inter-VLAN routing. Five access switches—SW-ADMIN, SW-EDIT, SW-SALES, SW-DESIGN, and SW-BOARD—are connected to SW-CORE. The SRV-FILE-PRINT-DNS server is connected directly to SW-CORE on VLAN 60, while the PC-MANAGEMENT host is connected to SW-CORE on VLAN 99. Each of the four departments has one end-user PC: PC-ADMIN, PC-EDIT, PC-SALES, and PC-DESIGN. The Boardroom is supported by SW-BOARD, which connects the BOARDROOM-WIRELESS-AP and BOARDROOM-WIRED-PRESENTATION devices.

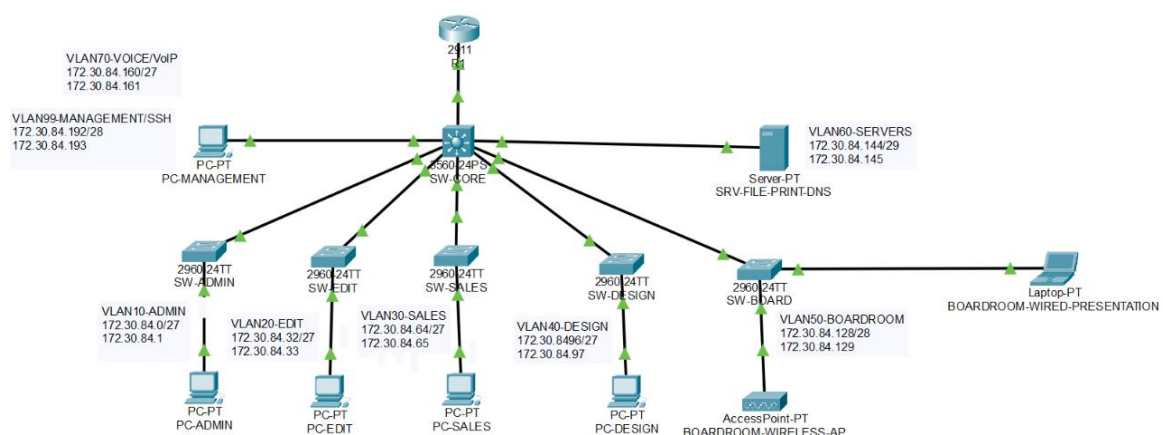


Figure 1 — Full network topology as built in Packet Tracer

3.2 VLAN Allocation

Eight VLANs were implemented in accordance with the allocation defined in Milestone 1. Each department is assigned a separate VLAN to provide network segmentation, while the servers have their own dedicated VLAN. Management traffic is kept separate from normal user traffic through VLAN 99. VLAN 70 has also been configured and assigned an IP address, but no access ports are currently connected to it because it has been reserved for the VoIP handsets specified in CR11.

Table 1 — VLAN allocation

VLAN	Name	Purpose	Access Ports
10	ADMIN	Administration & Management	SW-ADMIN Fa0/1–2
20	EDITORIAL	Editorial / Newsroom	SW-EDIT Fa0/1–2
30	SALES	Advertising & Sales	SW-SALES Fa0/1–2
40	DESIGN	Design & Pre-Press	SW-DESIGN Fa0/1–2
50	BOARDROOM	Boardroom (wired + wireless)	SW-BOARD Fa0/1–3
60	SERVERS	File / Print / DNS	SW-CORE Fa0/5
70	VOICE	VoIP handsets (CR11 reserve)	Reserved for future use
99	MANAGEMENT	SSH device management plane	SW-CORE Fa0/6

3.3 IP Addressing

The IP addressing follows the VLSM scheme established in Milestone 1 and is based on the allocated 172.30.84.0/23 address block. Subnet sizes were assigned according to the requirements of each VLAN, with /27 networks used for the departmental and voice VLANs, /28 networks for the Boardroom and management VLANs, and a /29 network for the server VLAN. The first usable IP address in each subnet is assigned as the default gateway through the corresponding SVI configured on SW-CORE.

Table 2 — IP addressing plan

VLAN / Link	Subnet	Mask	Gateway
10	172.30.84.0/27	255.255.255.224	172.30.84.1
20	172.30.84.32/27	255.255.255.224	172.30.84.33
30	172.30.84.64/27	255.255.255.224	172.30.84.65
40	172.30.84.96/27	255.255.255.224	172.30.84.97
50	172.30.84.128/28	255.255.255.240	172.30.84.129
60	172.30.84.144/29	255.255.255.248	172.30.84.145
70	172.30.84.160/27	255.255.255.224	172.30.84.161
99	172.30.84.192/28	255.255.255.240	172.30.84.193
R1 boundary link	172.30.84.208/30	255.255.255.252	172.30.84.209 (R1) / .210 (SW-CORE)

Change from Milestone 1: In Milestone 1, the 172.30.84.208/30 subnet was reserved and left unused. In the final implementation, it was configured as a point-to-point link between R1 (172.30.84.209) and SW-CORE (172.30.84.210). This link allows R1 to be accessed through SSH from the management VLAN and enables communication between R1 and SW-CORE. R1 remains isolated from the Internet and has no ISP connection.

3.4 Trunk Configuration

All inter-switch connections were configured as 802.1Q trunk links, with VLAN 99 set as the native VLAN on both ends of each connection. This ensures that untagged traffic is associated with the management VLAN instead of the default VLAN 1, while also keeping the native VLAN consistent across both ends of each trunk. An initial native VLAN mismatch caused errors during configuration, which is documented as Issue 1 in Section 7. Access-switch trunks were restricted to the required department VLAN and VLAN 99, while the SW-CORE trunk links were configured to allow all VLANs.

Table 3 — Trunk configuration

Parameter	Setting
Encapsulation	dot1q
Mode	trunk
Native VLAN	99 (both ends of every trunk)
Allowed VLANs — access switches	Department VLAN + VLAN 99
Allowed VLANs — SW-CORE trunks	All VLANs

3.5 Inter-VLAN Routing and SVI Configuration

SW-CORE handles all inter-VLAN routing through one SVI for each VLAN, with each SVI using the gateway address specified in Table 2. The access switches do not have SVIs configured for the data VLANs. Centralizing the routed interfaces on SW-CORE provides a single gateway for each VLAN and prevents duplicate addressing, which occurred when gateway IP addresses were incorrectly configured on the access switches (Issues 5–7). The access switches only use an SVI for VLAN 99 management.

Table 4 — SVI configuration on SW-CORE

Interface	VLAN	IP Address	Mask
Vlan10	ADMIN	172.30.84.1	255.255.255.224
Vlan20	EDITORIAL	172.30.84.33	255.255.255.224
Vlan30	SALES	172.30.84.65	255.255.255.224
Vlan40	DESIGN	172.30.84.97	255.255.255.224
Vlan50	BOARDROOM	172.30.84.129	255.255.255.240
Vlan60	SERVERS	172.30.84.145	255.255.255.248
Vlan70	VOICE	172.30.84.161	255.255.255.224
Vlan99	MANAGEMENT	172.30.84.193	255.255.255.240

```

SW-CORE
Physical Config CLI Attributes
IOS Command Line Interface
Fa0/3 10,20,30,40,50,60,70,99
Fa0/4 10,20,30,40,50,60,70,99
Gig0/2 10,20,30,40,50,60,70,99

Port Vlans allowed and active in management domain
Fa0/1 10,20,30,40,50,60,70,99
Fa0/2 10,20,30,40,50,60,70,99
Fa0/3 10,20,30,40,50,60,70,99
Fa0/4 10,20,30,40,50,60,70,99
Gig0/2 10,20,30,40,50,60,70,99

Port Vlans in spanning tree forwarding state and not pruned
Fa0/1 10,20,30,40,50,60,70,99
Fa0/2 10,20,30,40,50,60,70,99
Fa0/3 10,20,30,40,50,60,70,99
Fa0/4 10,20,30,40,50,60,70,99
Gig0/2 10,20,30,40,50,60,70,99

SW-CORE#show vlan brief

VLAN Name      Status      Ports
-----
1    default      active      Fa0/7, Fa0/8, Fa0/9, Fa0/10
          Fa0/11, Fa0/12, Fa0/13, Fa0/14
          Fa0/15, Fa0/16, Fa0/17, Fa0/18
          Fa0/19, Fa0/20, Fa0/21, Fa0/22
          Fa0/23, Fa0/24

10   ADMIN        active
20   EDITORIAL    active
30   SALES        active
40   DESIGN       active
50   BOARDROOM    active
60   SERVERS      active      Fa0/5
70   VOICE        active
99   MANAGEMENT   active      Fa0/6

1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
SW-CORE#write memory
Building configuration...
[OK]

```

Figure 2 — VLAN list on SW-CORE (show vlan brief)

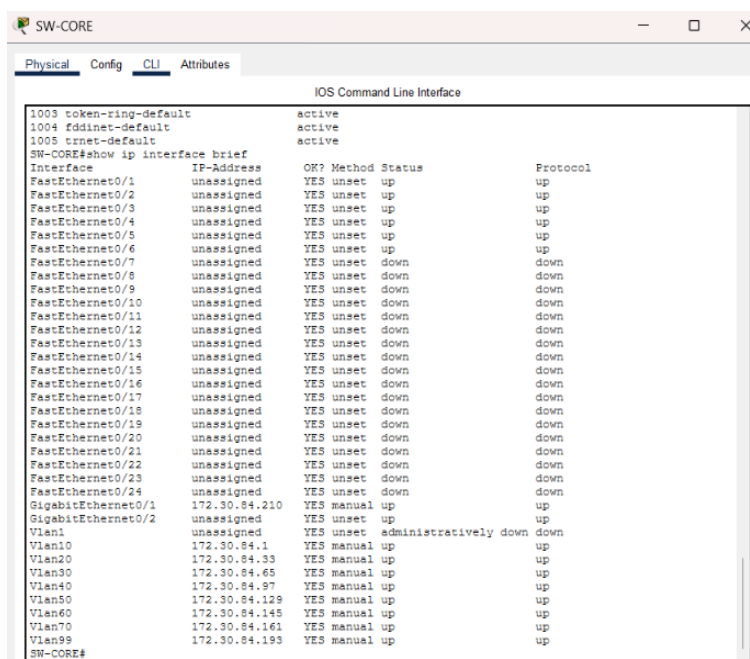


Figure 3 — SVI status on SW-CORE (show ip interface brief)

3.6 Management Plane Addressing

Every device is reachable for management on VLAN 99. SW-CORE holds the gateway address .193, and each device and the management host has a unique address in the same subnet.

Table 5 — Management plane IP addresses (VLAN 99)

Device	Management IP
SW-CORE (VLAN 99 SVI)	172.30.84.193 (gateway)
PC-MANAGEMENT	172.30.84.194
SW-ADMIN	172.30.84.195
SW-EDIT	172.30.84.196
SW-SALES	172.30.84.197
SW-DESIGN	172.30.84.198
SW-BOARD	172.30.84.199

3.7 Router Configuration (R1)

R1 functions as an internal boundary router. Its Gig0/0 interface is connected to SW-CORE through the /30 point-to-point link, while a default route directs unknown traffic towards SW-CORE. A specific static route to the management subnet was also configured to allow R1 to reach VLAN 99, as documented in Issue 4. Since the network does not have an external connection, the Gig0/1 and Gig0/2 uplink interfaces remain administratively shut down.

Table 6 — R1 configuration summary

Item	Configuration
Gig0/0	172.30.84.209/30
Default route	ip route 0.0.0.0 0.0.0.0 172.30.84.210
Management subnet route	ip route 172.30.84.192 255.255.255.240 172.30.84.210
Gig0/1, Gig0/2	Administratively down (no uplinks)

4. Assigned Networking Feature: SSH

4.1 Design Approach

The assigned challenge is the implementation of a secure device management plane using Secure Shell (SSH). Since Telnet transmits credentials and commands in clear text, SSH was used to provide encrypted remote management, with device administration restricted to SSH connections. Management access is further separated from departmental user traffic through dedicated VLAN 99, which contains the management addresses for the network devices. Administration is performed from PC-MANAGEMENT within VLAN 99 and from R1 through the boundary link.

4.2 Configuration Applied

The same SSH configuration was applied to all seven devices: R1, SW-CORE, SW-ADMIN, SW-EDIT, SW-SALES, SW-DESIGN and SW-BOARD.

Listing 1 — SSH configuration applied to all devices

```
ip domain-name molemane.local
username admin privilege 15 secret Cisco123!
crypto key generate rsa general-keys modulus 2048
line vty 0 4
  transport input ssh
  login local
exit
ip ssh version 2
```

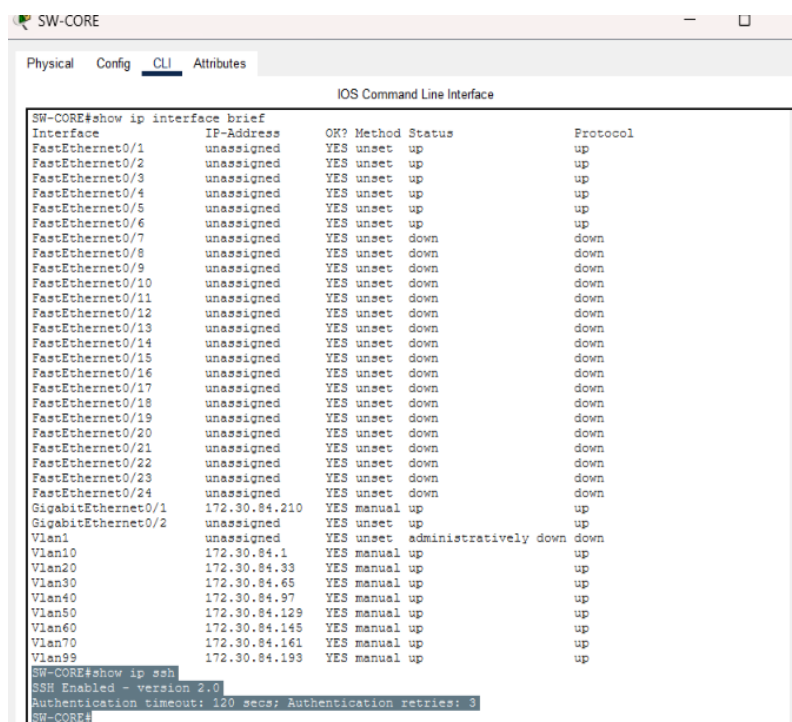
4.3 Key Decisions

- **Domain name and RSA keys:** A domain name was configured before generating the RSA keys. A 2048-bit modulus was selected to provide a suitable level of security supported by the network devices.
- **Local authentication:** A local administrator account with privilege level 15 and a hashed secret was configured to provide authentication without relying on an external authentication server. Centralized AAA authentication is outside the scope of this challenge.
- **SSH only on the VTY lines:** The VTY lines were configured with transport input ssh, preventing Telnet connections on VTY lines 0–4. The login local command ensures that the configured local account is used for authentication.
- **SSH version 2:** SSH version 2 was explicitly enabled to ensure that the older SSH version 1 protocol is not used.

- **VLAN 99 as the management and native VLAN:** VLAN 99 was designated as the management VLAN and native VLAN. This separates management traffic from VLAN 1, the default VLAN, and provides a consistent VLAN for device management.
- **No data-VLAN SVIs on access switches:** Each access switch has only a VLAN 99 management SVI. This ensures that SW-CORE remains the only device providing the routed gateway for the user VLANs.

4.4 Verification

SSH was verified using the show ip ssh command on SW-CORE, which confirmed that SSH was enabled and operating with version 2.0. SSH connections to SW-CORE at 172.30.84.193 using the admin account were successfully established from both PC-MANAGEMENT and R1. Further verification results are presented in Section 5.2.



```

SW-CORE#show ip interface brief
Interface      IP-Address      OK? Method Status  Protocol
FastEthernet0/1 unassigned     YES unset  up      up
FastEthernet0/2 unassigned     YES unset  up      up
FastEthernet0/3 unassigned     YES unset  up      up
FastEthernet0/4 unassigned     YES unset  up      up
FastEthernet0/5 unassigned     YES unset  up      up
FastEthernet0/6 unassigned     YES unset  up      up
FastEthernet0/7 unassigned     YES unset  down    down
FastEthernet0/8 unassigned     YES unset  down    down
FastEthernet0/9 unassigned     YES unset  down    down
FastEthernet0/10 unassigned     YES unset  down    down
FastEthernet0/11 unassigned     YES unset  down    down
FastEthernet0/12 unassigned     YES unset  down    down
FastEthernet0/13 unassigned     YES unset  down    down
FastEthernet0/14 unassigned     YES unset  down    down
FastEthernet0/15 unassigned     YES unset  down    down
FastEthernet0/16 unassigned     YES unset  down    down
FastEthernet0/17 unassigned     YES unset  down    down
FastEthernet0/18 unassigned     YES unset  down    down
FastEthernet0/19 unassigned     YES unset  down    down
FastEthernet0/20 unassigned     YES unset  down    down
FastEthernet0/21 unassigned     YES unset  down    down
FastEthernet0/22 unassigned     YES unset  down    down
FastEthernet0/23 unassigned     YES unset  down    down
FastEthernet0/24 unassigned     YES unset  down    down
GigabitEthernet0/1 172.30.84.210 YES manual up      up
GigabitEthernet0/2 unassigned     YES unset  up      up
Vlan1             unassigned     YES unset  administratively down down
Vlan10             172.30.84.1   YES manual up      up
Vlan20             172.30.84.33  YES manual up      up
Vlan30             172.30.84.65  YES manual up      up
Vlan40             172.30.84.97  YES manual up      up
Vlan50             172.30.84.129 YES manual up      up
Vlan60             172.30.84.145 YES manual up      up
Vlan70             172.30.84.161 YES manual up      up
Vlan99            172.30.84.193 YES manual up      up

SW-CORE#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
SW-CORE#

```

Figure 4 — SSH version 2 enabled on SW-CORE

```
PC-MANAGEMENT
Physical Config Desktop Programming Attributes
Command Prompt
[Connection to 172.30.84.194 closed by foreign host]
C:\>ssh -l admin 172.30.84.194
Password:
% Password: timeout expired!
% Login invalid
[Connection to 172.30.84.194 closed by foreign host]
C:\>ssh -l admin 172.30.84.194
Password:
SW-CORE#show run | include hostname
hostname SW-CORE
SW-CORE#exit
[Connection to 172.30.84.194 closed by foreign host]
C:\>telnet 172.30.84.194
Trying 172.30.84.194 ...Open
[Connection to 172.30.84.194 closed by foreign host]
C:\>ping 172.30.84.193
Pinging 172.30.84.193 with 32 bytes of data:
Reply from 172.30.84.193: bytes=32 time<1ms TTL=255
Reply from 172.30.84.193: bytes=32 time<1ms TTL=255
Reply from 172.30.84.193: bytes=32 time<1ms TTL=255
Reply from 172.30.84.193: bytes=32 time<1ms TTL=255
Ping statistics for 172.30.84.193:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ssh -l admin 172.30.84.193
Password:
SW-CORE#
```

Figure 5 — Successful SSH login from PC-MANAGEMENT to SW-CORE

4.5 Telnet Rejection

A Telnet connection was attempted from PC-MANAGEMENT to SW-CORE using telnet 172.30.84.193. The connection was closed by the remote host, and no login session was established. This confirmed that Telnet access is not permitted and that SSH is the only protocol allowed for remote device management.

```
PC-MANAGEMENT
Physical Config Desktop Programming Attributes
Command Prompt
[Connection to 172.30.84.194 closed by foreign host]
C:\>ssh -l admin 172.30.84.194
Password:
SW-CORE#show run | include hostname
hostname SW-CORE
SW-CORE#exit
[Connection to 172.30.84.194 closed by foreign host]
C:\>telnet 172.30.84.194
Trying 172.30.84.194 ...Open
[Connection to 172.30.84.194 closed by foreign host]
C:\>ping 172.30.84.193
Pinging 172.30.84.193 with 32 bytes of data:
Reply from 172.30.84.193: bytes=32 time<1ms TTL=255
Reply from 172.30.84.193: bytes=32 time<1ms TTL=255
Reply from 172.30.84.193: bytes=32 time<1ms TTL=255
Reply from 172.30.84.193: bytes=32 time<1ms TTL=255
Ping statistics for 172.30.84.193:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ssh -l admin 172.30.84.193
Password:
SW-CORE#exit
[Connection to 172.30.84.193 closed by foreign host]
C:\>telnet 172.30.84.193
Trying 172.30.84.193 ...Open
[Connection to 172.30.84.193 closed by foreign host]
C:\>
```

Figure 6 — Telnet connection rejected by SW-CORE

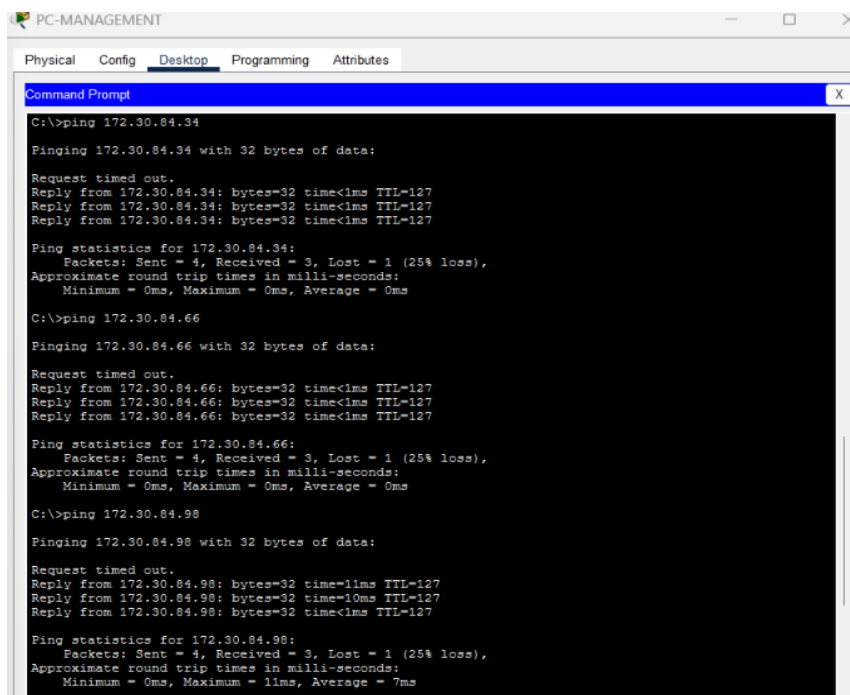
5. Testing and Verification

5.1 Inter-VLAN Connectivity Tests

Inter-VLAN routing was tested by pinging from PC-ADMIN (172.30.84.2, VLAN 10) to a host in each other VLAN. Each test sent four packets.

Table 7 — Inter-VLAN ping tests from PC-ADMIN

Destination	IP	VLAN	Sent	Received	Lost	Status
PC-EDIT	172.30.84.34	20	4	3	25%	Success
PC-SALES	172.30.84.66	30	4	3	25%	Success
PC-DESIGN	172.30.84.98	40	4	3	25%	Success
Server	172.30.84.146	60	4	3	25%	Success
Boardroom Laptop	172.30.84.130	50	4	3	25%	Success



```
PC-MANAGEMENT
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping 172.30.84.34
Pinging 172.30.84.34 with 32 bytes of data:
Request timed out.
Reply from 172.30.84.34: bytes=32 time<1ms TTL=127
Reply from 172.30.84.34: bytes=32 time<1ms TTL=127
Reply from 172.30.84.34: bytes=32 time<1ms TTL=127
Ping statistics for 172.30.84.34:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 172.30.84.66
Pinging 172.30.84.66 with 32 bytes of data:
Request timed out.
Reply from 172.30.84.66: bytes=32 time<1ms TTL=127
Reply from 172.30.84.66: bytes=32 time<1ms TTL=127
Reply from 172.30.84.66: bytes=32 time<1ms TTL=127
Ping statistics for 172.30.84.66:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 172.30.84.98
Pinging 172.30.84.98 with 32 bytes of data:
Request timed out.
Reply from 172.30.84.98: bytes=32 time=11ms TTL=127
Reply from 172.30.84.98: bytes=32 time=10ms TTL=127
Reply from 172.30.84.98: bytes=32 time<1ms TTL=127
Ping statistics for 172.30.84.98:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 7ms
```

Figure 7 — Inter-VLAN ping tests (part 1)

```

C:\>ping 172.30.84.146

Pinging 172.30.84.146 with 32 bytes of data:

Request timed out.
Reply from 172.30.84.146: bytes=32 time<1ms TTL=127
Reply from 172.30.84.146: bytes=32 time<1ms TTL=127
Reply from 172.30.84.146: bytes=32 time<1ms TTL=127

Ping statistics for 172.30.84.146:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 172.30.84.130

Pinging 172.30.84.130 with 32 bytes of data:

Request timed out.
Reply from 172.30.84.130: bytes=32 time<1ms TTL=127
Reply from 172.30.84.130: bytes=32 time<1ms TTL=127
Reply from 172.30.84.130: bytes=32 time<1ms TTL=127

Ping statistics for 172.30.84.130:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

```

Figure 8 — Inter-VLAN ping tests (part 2)

5.2 SSH Tests

Table 8 — SSH verification

Test	Target	Result
show ip ssh on SW-CORE	SW-CORE	SSH Enabled — version 2.0
SSH login as admin from PC-MANAGEMENT	SW-CORE (172.30.84.193)	Successful
SSH login from R1	SW-CORE	Successful
Telnet from PC-MANAGEMENT	SW-CORE (172.30.84.193)	Rejected (connection closed)

5.3 Trunk Tests

Trunk operation was verified using the show interfaces trunk command on SW-CORE, which displayed the trunk interfaces, encapsulation type, native VLAN, and allowed VLANs. The initial native VLAN mismatch was resolved by configuring VLAN 99 as the native VLAN on both ends of each trunk, as documented in Issue 1.

```

SW-CORE#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3

SW-CORE#show interfaces trunk

```

Port	Mode	Encapsulation	Status	Native VLAN
Fa0/1	on	802.1q	trunking	99
Fa0/2	on	802.1q	trunking	99
Fa0/3	on	802.1q	trunking	99
Fa0/4	on	802.1q	trunking	99
Gig0/2	on	802.1q	trunking	99

```

Port      Vlans allowed on trunk
Fa0/1     10,20,30,40,50,60,70,99
Fa0/2     10,20,30,40,50,60,70,99
Fa0/3     10,20,30,40,50,60,70,99
Fa0/4     10,20,30,40,50,60,70,99
Gig0/2    10,20,30,40,50,60,70,99

Port      Vlans allowed and active in management domain
Fa0/1     10,20,30,40,50,60,70,99
Fa0/2     10,20,30,40,50,60,70,99
Fa0/3     10,20,30,40,50,60,70,99
Fa0/4     10,20,30,40,50,60,70,99
Gig0/2    10,20,30,40,50,60,70,99

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     10,20,30,40,50,60,70,99
Fa0/2     10,20,30,40,50,60,70,99
Fa0/3     10,20,30,40,50,60,70,99
Fa0/4     10,20,30,40,50,60,70,99
Gig0/2    10,20,30,40,50,60,70,99

SW-CORE#

```

Figure 9 — Trunk status on SW-CORE (show interfaces trunk)

6. Interpretation of Results

The inter-VLAN tests consistently returned three successful replies out of four packets, resulting in 25% packet loss. This behavior occurred during the initial packet exchange and was attributed to ARP resolution rather than a routing fault. The first ICMP packet may time out while the source device resolves the MAC address of its default gateway and the gateway resolves the destination address. Subsequent packets were successfully delivered, confirming that inter-VLAN routing between VLAN 10 and VLANs 20, 30, 40, 50, and 60 through the SVIs on SW-CORE was functioning correctly.

The SSH test confirmed that the management plane operated as intended. Successful SSH sessions were established from PC-MANAGEMENT, which is located in VLAN 99, as well as from R1, which reaches the management subnet through the static route configured during troubleshooting. The Telnet test resulted in the connection being immediately closed by the remote device without displaying a login prompt. This behavior confirmed that the VTY lines were restricted to SSH through the transport input ssh configuration.

VLAN 70 was not included in the testing because it is reserved for the future CR11 VoIP requirement and currently has no connected hosts. Its SVI has nevertheless been configured, allowing voice devices to be added in the future without requiring changes to the existing routing design.

7. Troubleshooting Log

Several issues were identified and resolved during the implementation process. These issues are documented in Table 9 and demonstrate that the final network configuration was developed iteratively, with each stage being tested and verified before proceeding to the next.

Table 9 — Issues identified and resolved

#	Issue	Root Cause	Resolution
1	Native VLAN mismatch errors on trunks	Access switches defaulted to VLAN 1 native	Set native VLAN 99 on both ends of all trunks
2	Missing VLAN 50, 60, 70, 99 SVIs on SW-CORE	Initial SVI config stopped after VLAN 40	Added missing SVIs with correct IPs
3	Ping to Server failed	VLAN 60 SVI missing	Added interface vlan 60 with 172.30.84.145/29
4	SSH login timeout from R1	No route from R1 to VLAN 99 subnet	Added static route to 172.30.84.192/28 via SW-CORE
5–7	Duplicate IP on SW-EDIT / SW-SALES / SW-DESIGN	Data-VLAN SVIs were given gateway IPs	Removed IP and shut down data-VLAN SVIs on access switches
8	Fa0/1 on SW-BOARD still in default VLAN	Port not reassigned after cable move	Set Fa0/1 to access VLAN 50
9	VLAN 99 IP conflict (SW-CORE and PC-MANAGEMENT)	Both initially assigned .194	SW-CORE → .193 (gateway); PC-MANAGEMENT → .194 (host)

8. Conclusion

The implementation successfully reflects the network design established in Milestone 1. Departmental segmentation is provided through VLANs 10–40, while VLAN 50 is dedicated to the Boardroom and supports both wired and wireless connectivity. The file, print and DNS server is placed in VLAN 60, with SW-CORE responsible for inter-VLAN routing. VLAN 70 has been configured and reserved for the VoIP handsets specified in CR11, allowing future voice services to be introduced without redesigning the network. The assigned SSH challenge was implemented across all seven network devices, with Telnet access disabled and device management restricted to VLAN 99.

The main change from the Milestone 1 design is the use of the previously reserved 172.30.84.208/30 subnet as the point-to-point link between R1 and SW-CORE. The troubleshooting log documents the configuration issues encountered, and the corrections made during implementation. The network is now functionally ready for final submission, which includes the Packet Tracer file, GitHub portfolio, technical report, and demonstration video.

9. References

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